

# **Lab 3 – Architecture Selection Justification**

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**Problem Statement: Household Carbon Footprint & Target Tracker System**

## **Architecture Selection**

We chose Layered Architecture for the Household Carbon Footprint & Target Tracker System.

## **Two Reasons**

1. The system has a natural separation between the user interface, business functions such as CO<sub>2</sub>e calculation and target tracking, and stored household data.
2. Keeping these responsibilities in separate layers makes the system easier to understand, test and maintain for the scope of the project.

## **Security Advantage**

Household consumption and emissions can be kept behind the business and data layers. Authorization can be checked before protected records are returned, limiting access to the authorized Resident and assigned Sustainability Coach.

## **Performance Benefit**

Separating calculation and data access keeps each responsibility focused and reduces unnecessary work in the presentation layer. This supports the requirement that the annual reduction milestone chart render in under 200 ms.

## **Components Used**

Resident UI, Coach UI, Consumption Manager, CO<sub>2</sub>e Calculation Service, Target & Progress Service, Household Data Repository and the external Emission Factor Service.